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Question Paper Code : 19314

B.E / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2022

Third Semester

Common to B.E. – Computer Science and Engineering & B.Tech. Information Technology

19CS302 – DIGITAL PRINCIPLES AND SYSTEM DESIGN

(Regulations: Mepco – R2019)

Duration: 3 Hours | Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

A,CO1 1. The excess-3 code for 597 is given by _____. Justify your answer.

A) 100011001010 B) 100010100111

C) 010110010111 D) 010110101101

A,CO1 2. Which one of the following expressions does NOT represent exclusive NOR of x and y? Justify your answer.

A) xy+x'y' B) x ⊕ y' C) x'⊕ y D) x'⊕ y'

A,CO2 3. How many 3-to-8 line decoders with an enable input are needed to construct a 6-to-64 line decoder without using any other logic gates? Justify your answer.

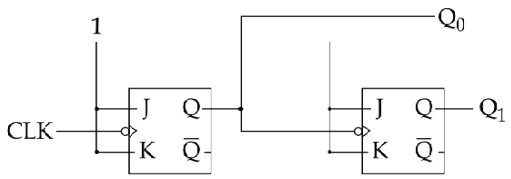
A) 7 B) 8 C) 9 D) 10

U,CO2 4. Compare and contrast Combinational Circuits with Sequential Circuits.

U,CO3 5. Derive the expression for Sum and Carry of a Full Adder.

A,CO3 6. Construct a 3-bit Johnson Counter using D flips flop.

A,CO4 7. What does the following logic diagram represent? Justify your answer.



A) Synchronous Counter B) Ripple Counter

C) Combinational Circuit D) Mod 2 Counter

A,CO4 8. Characteristic equation of an SR flip-flop is given by _____. Justify your answer.

A) $Q_{n+1} = S + RQ_n$ B) $Q_{n+1} = RQ'_n + SQ_n$

C) $Q_{n+1} = S' + RQ_n$ D) $Q_{n+1} = S + R'Q_n$

U,CO5 9. How truth table differs from flow and transition table?

R,CO5 10. What do you mean by Hazards?

PART B – (5 × 16 = 80 Marks)

A,CO1 11. a) i. Simplify the Boolean function f (A, B, C, D) = Σ (1, 4, 5, 9, 11, 12) in Product of Sums format using K-map. (8 Marks)

A,CO1 11. a) ii. Simplify the Boolean expressions using Algebraic method:

A) x'z+w'x y'+w(x'y+xy')

B) A'B'C'+B'CD'+A'BCD'+AB'C' (8 Marks)

OR

A,CO1 11. b) Simplify the expression F(A,B,C,D) = ∑(0,6,8,13,14) + $\sum_d$(2,4,10) using Quine-McClusky – Tabulation method and implement the circuit using NOR gates. (16 Marks)

A, ,CO2 12. a) Design a combinational circuit for 4 bit Gray code to Binary code converter and write the HDL behavioral description of the same. (16 Marks)

OR

A, ,CO2 12. b) i. Implement the following Boolean function using Multiplexer.

F(A,B,C,D) = ∑(1,3,4,11,12,13,14,15) (8 Marks)

A. ,CO2 12. b) ii. Construct Two-to-Four–line decoder with enable input using NAND and NOT gates. (8 Marks)

A,CO3 13. a) i. Implement the following functions using Programmable Logic Array. Construct Truth table, Program table, K-map simplification and draw the PLA circuit.

F1(x, y, z) = ∑(1, 2,4,6)

F2 (x, y, z) = ∑(0, 1, 6, 7)

F3(x, y, z) = ∑(2,6) (8 Marks)

U,CO3	13. a) ii.	Write notes on classification of memories.	(8 Marks)
		OR	
A,CO3	13. b) i.	Design a combinational circuit using ROM, the circuit accepts a three-bit number and outputs a binary number equal to the square of the input number.	(8 Marks)
U,CO3	13. b) ii.	Explain how Coincident decoding can reduce the number of gates in a memory decoder.	(8 Marks)
U,CO4	14. a)	Explain the principle of a BCD ripple counter with a neat logic diagram and state diagram.	(16 Marks)
		OR	
U,CO4	14. b) i.	Compare the diagram of 4 bit SISO, SIPO, PIPO and PIPO shift registers and draw its waveforms.	(8 Marks)
U,CO4	14. b) ii.	Realize D flip-flop using SR flip-flop.	(8 Marks)
U,CO5	15. a) i.	Explain race-free state assignment with an example.	(8 Marks)
A,CO5	15. a) ii.	Design an asynchronous sequential circuit that has two inputs X2 and X1 and one output Z. The output is to remain 0 as long as X1 is 0. The first change in X2 that occurs while X1 is 1 will cause Z to be 1. Z is to remain 1 until X1 returns to 0. Construct state diagram and flow table.	(8 Marks)
		OR	
U,CO5	15. b) i.	Compare and Contrast Synchronous and Asynchronous sequential circuit and also draw the block diagram of an asynchronous circuit by explaining the designing steps.	(8 Marks)
A ,CO5	15. b) ii.	Describe the different types of Hazards. Also, eliminate hazard from the following function: $F = \pi(1, 2, 3, 6, 14)$	(8 Marks)